

HARBOR SPRINGS, MI, USA

WMO: 722145

Lat: 45.426N

Lon: 84.913W

Elev: 677

StdP: 14.34

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 04884

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.9	2.6	-9.6	3.4	-1.6	-4.3	4.5	4.6	23.7	28.0	20.7	26.6	3.3	50	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	83.0	69.1	81.1	67.9	79.0	66.8	72.6	79.3	71.1	77.5	69.5	75.7	6.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.4	114.8	76.3	68.9	109.0	75.3	67.4	103.4	73.7	36.7	79.2	35.4	77.4	34.0	76.5	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.7	17.2	15.3	-10.4	90.0	7.8	1.5	-16.0	91.0	-20.5	91.9	-24.9	92.7	-30.5	93.8		
(5)			-10.9	75.7	7.6	1.9	-16.4	77.1	-20.8	78.2	-25.1	79.3	-30.6	80.7	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	45.2	22.1	21.6	30.0	41.3	52.7	62.1	67.7	67.1	61.0	49.0	37.8	28.1	(6)
(7)		DBStd	18.49	10.17	10.01	11.00	8.86	8.42	6.71	5.29	5.16	7.30	8.26	8.46	8.44	(7)
(8)		HDD50	3822	864	796	627	287	65	2	0	0	5	121	376	679	(8)
(9)		HDD65	7554	1329	1216	1085	710	393	130	33	35	162	501	816	1144	(9)
(10)		CDD50	2062	0	0	7	26	149	366	550	531	334	90	9	0	(10)
(11)		CDD65	322	0	0	0	0	13	44	118	101	41	5	0	0	(11)
(12)		CDH74	2056	0	0	3	6	115	312	754	591	247	28	0	0	(12)
(13)		CDH80	321	0	0	0	0	20	53	121	89	36	2	0	0	(13)
(14)	Wind	WSAvg	6.0	7.0	6.9	6.1	6.9	5.6	4.8	4.3	4.6	5.0	6.5	6.8	7.5	(14)
(15)	Precipitation	PrecAvg	31.4	2.2	1.5	1.6	2.3	2.9	3.2	2.7	3.3	3.5	3.6	2.8	2.4	(15)
(16)		PrecMax	36.7	4.1	2.4	5.4	4.1	5.2	7.0	5.0	6.0	6.5	7.8	5.1	4.1	(16)
(17)		PrecMin	26.2	0.6	0.4	0.0	0.8	0.7	0.9	1.4	0.8	1.1	1.1	1.1	0.3	(17)
(18)		PrecStd	3.0	0.9	0.6	1.1	0.9	1.1	1.6	1.0	1.4	1.2	1.7	1.1	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.5	48.2	68.7	73.5	82.2	86.1	88.4	87.6	84.2	78.6	66.0	54.2	(19)
(20)			MCWB	43.2	43.6	57.5	55.8	65.4	69.7	72.2	72.0	69.6	65.4	56.1	51.2	(20)
(21)		2%	DB	41.2	41.2	56.0	67.6	77.1	81.0	82.4	82.0	80.5	72.5	58.5	44.9	(21)
(22)			MCWB	38.1	37.4	48.9	53.4	63.5	66.7	69.1	69.3	67.9	63.5	51.7	41.9	(22)
(23)		5%	DB	37.5	37.0	50.1	61.0	71.9	77.7	80.9	79.8	76.6	67.9	54.1	41.1	(23)
(24)			MCWB	35.1	34.0	44.1	49.0	59.6	65.3	68.0	67.6	66.3	60.6	48.1	38.3	(24)
(25)		10%	DB	35.7	34.7	43.4	54.9	67.6	74.6	78.7	77.1	73.4	61.9	50.2	38.5	(25)
(26)			MCWB	33.7	32.1	37.6	45.4	58.2	64.0	67.2	67.2	65.1	55.5	45.8	36.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.3	44.6	56.7	60.0	69.0	72.9	75.2	74.1	72.9	69.6	57.3	51.5	(27)
(28)			MCDWB	45.8	48.0	66.0	69.4	79.0	81.4	83.5	80.9	79.2	75.2	64.1	53.5	(28)
(29)		2%	WB	38.4	37.7	49.7	54.5	65.9	69.5	72.6	72.5	70.7	65.5	52.6	42.2	(29)
(30)			MCDWB	40.9	40.7	55.0	64.3	73.8	77.0	79.1	79.0	76.2	70.0	57.4	44.2	(30)
(31)		5%	WB	35.5	34.5	43.6	50.2	62.4	67.7	71.0	70.9	68.7	60.3	49.1	38.8	(31)
(32)			MCDWB	37.3	36.7	50.6	59.5	70.2	75.0	77.4	76.9	74.6	66.5	53.5	40.7	(32)
(33)		10%	WB	33.4	32.4	38.5	46.4	58.5	65.8	69.2	69.1	66.5	56.5	46.1	36.5	(33)
(34)			MCDWB	35.1	34.3	42.7	53.8	65.3	72.2	75.7	74.9	72.4	61.7	49.5	38.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.4	13.5	16.2	17.3	19.6	18.3	17.8	17.3	18.2	14.7	11.6	9.5	(35)
(36)			MCDBR	11.5	14.3	22.0	27.3	26.3	23.0	20.9	20.0	20.8	21.5	18.4	12.8	(36)
(37)			MCWBR	10.0	12.1	15.0	16.0	14.6	12.1	10.8	10.7	11.4	13.7	13.0	11.5	(37)
(38)		5% WB	MCDWB	10.8	13.5	21.2	25.4	23.3	18.5	16.7	15.6	17.4	19.5	17.4	12.1	(38)
(39)			MCWBR	9.9	12.0	15.1	16.1	14.7	11.5	9.9	9.7	11.6	14.1	13.3	11.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.297	0.292	0.316	0.353	0.377	0.392	0.389	0.382	0.369	0.337	0.303	0.278	(40)	
(41)		taud	2.315	2.324	2.385	2.347	2.354	2.342	2.373	2.390	2.413	2.490	2.518	2.446	(41)	
(42)		Ebn at Noon	255	282	291	287	283	278	277	275	269	264	256	254	(42)	
(43)		Edh at Noon	25	30	32	37	38	39	37	35	32	25	21	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	369	663	1125	1467	1793	1948	1960	1665	1272	741	394	271	(44)	
(45)		RadStd	41	67	116	150	153	141	73	102	106	86	56	27	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (40 neighbors)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		-2.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page